

Hamdi Barkous

🌐 hamdibarkous.com — 📍 Hamdi Barkous — ✉ hamdi.barkous@ept.ucar.tn — in HamdiBarkous

PROFESSIONAL SUMMARY

Machine Learning Engineer with expertise in developing and deploying AI solutions for real-world applications. Proven track record in building sophisticated deep learning models, implementing agentic frameworks, and creating production-ready systems. Leverages cutting-edge AI development tools to rapidly prototype and deliver innovative solutions that blend advanced algorithms with practical implementation. Experienced in financial modeling, time series forecasting, and full-stack AI applications.

EXPERIENCE

FinGenesis - ML Engineer

2024 – present

- Designed and implemented DL&ML models to predict price movements of financial symbols.
- Developed an agentic framework for ticker-based sentiment analysis, enabling actionable insights from financial data.
- Designed and implemented rule-based trading strategies.
- Designed and implemented LLM-based trading strategies with explainability as a key feature.
- Designed a backtesting framework for evaluating trading strategies.
- Optimized data pipeline efficiency, reducing processing time by 60% and enhancing performance.

PROJECTS

IntoTeX - AI-Powered PDF to LaTeX Converter – [🔗](#)

In Development

Keywords: Full-Stack SaaS, AI/LLM, LangGraph, Vision AI, WebSocket, Payments

- Built a full-stack SaaS platform that converts PDF documents to LaTeX source code using a LangGraph-based AI agent with vision capabilities
- Developed a sophisticated PDF extraction engine capable of parsing complex document structures including formatted text, raster images, and embedded vector graphics
- Developed a credit-based subscription system with Paddle payment integration

T3 Chat Clone - AI Chatbot with MCP Tools Integration – [🔗](#) [📄](#)

Live Production

Keywords: Full-Stack Development, AI/LLM, MCP Tools, Real-time Streaming, Multi-modal AI

- Built and deployed a full-stack AI chatbot with MCP (Model Context Protocol) tools integration and multi-modal input support
- Implemented real-time streaming, chat branching, and full authentication system
- Deployed to production with AWS backend and Vercel frontend infrastructure

PUBLICATIONS

Barkous, H., Amayri, M., & Bouguila, N. “**A Comprehensive Analysis of a Hybrid Deep Learning Model for Midterm Electric Load Forecasting**” 21st IEEE International Conference on Smart Cities, Australia, Dec 13, 2023

EDUCATION

École Polytechnique de Tunisie – ABET Accredited

2020 – 2023

Multidisciplinary Engineering - Minor: Signals & systems

Relevant Courses:

- **Mathematics:** Optimization, Detection and Estimation, Probability & Statistics, Stochastic Processes
- **Computer Science:** Artificial Intelligence, Data Science, Data Analysis, Algorithms and Data Structures.
- **Signals and Systems:** Signal Processing, Digital Communication, Control Theory, Information Theory

TECHNICAL SKILLS

- **Programming Languages:** Python, TypeScript, SQL
- **ML/AI Frameworks:** LangChain, LangGraph, PyTorch, TensorFlow, OpenCV
- **API Development Stack:** FastAPI, Next.js, PostgreSQL, Redis, Supabase
- **ML Platforms & Tools:** Hugging Face, MLflow
- **DevOps & Cloud:** AWS, Vercel, Docker, Git
- **Developer Tools:** Cursor, MCP